

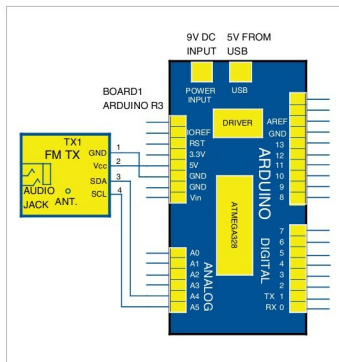


Fig. 4: Circuit diagram of FM station



Fig. 6: Author's prototype

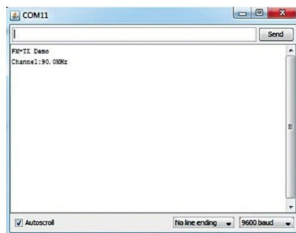


Fig. 5: Serial Monitor window

different frequency from Serial Monitor by simply entering the frequency using & symbol. For example, to change frequency to 99MHz type &0990 in the Serial Monitor and press Enter key.

Similarly, to change frequency to 99.5MHz type &0995 (for frequency 101MHz type &1010 and for 102.5MHz type &1025) in the Serial Monitor.

Software

Circuit operation is done using the software (fmtx_demo.ino) loaded into the internal memory of Arduino Uno R3. The program implements all required functionalities includ-

ing handling user inputs with the help of serial port interface. The program is written in Arduino programming language.

Arduino IDE 1.6.4 is used to compile and upload the program to Arduino board. The important thing is to set any frequency (within FM band), as shown in Fig. 5. In the program, default frequency is set to 90.1MHz, as given below:

```
float fm_freq = 90.1;
```

When you apply 9V power supply to Arduino board, it starts broadcasting the audio signal at 90.1MHz frequency.

You can simply tune to the same frequency in the FM receiver or mobile phone. You can now listen to music or message announcements from up to a distance of 50 metres.

Note that, an external antenna is required for FM transmitter module.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Connect any metal wire to the pad that could serve as an antenna. Theoretically, the ideal antenna length for 100MHz is around 75cm.

Construction and testing

You do not need a PCB for this project. Connect GND, 5V, A4 and A5 pins of Arduino board to GND, Vcc, SDA and SLA pins of FM transmitter module, respectively, using external wires/jumpers.

Connect a USB cable to Arduino Uno R3 for compiling and uploading the program. Remove the cable after uploading it. Connect a 9V battery/adaptor as power supply to Arduino Uno R3.

Connect FM transmitter V2.0 module to any audio source like MP3 player/mobile phone/computer using a 3.5mm audio cable. Your circuit will transmit at 90.1MHz frequency, which can be received by any FM radio by tuning to the same frequency. The author's prototype is shown in Fig. 6. **EFY**



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